Matthias Rottner

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Employment

08/2026 - present	Senior Economist, Deutsche Bundesbank, Research Centre
07/2024 - 07/2026	Economist, Bank for International Settlements (BIS)
05/2021 - 07/2024	Economist, Deutsche Bundesbank, Research Centre (on leave 07/2024 - 07/2026)

Education

09/2016 - 05/2021	Ph.D. in Economics, European University Institute Advisors: Evi Pappa, Leonardo Melosi
09/2016 - 08/2017	MRes in Economics, European University Institute
09/2014 - 08/2016	MSc in Economics, University of Copenhagen
04/2011 - 03/2014	BA in Economics, University of Erlangen-Nürnberg

Research and Professional Experience

02/2023 - 03/2023	Visiting Researcher, Bank of Estonia
09/2020 - 04/2021	Consultant, European Central Bank
09/2019 - 08/2020	PhD Traineeship, European Central Bank
01/2019 - 06/2019	Visiting Researcher, Federal Reserve Bank of Chicago
01/2019 - 06/2019	Visiting Researcher, Northwestern University
08/2018 - 12/2018	Internship, Deutsche Bundesbank
Summer 2015,16,17	Internship, Bank of Estonia
04/2014 - 06/2014	Internship, Kiel Institute for the World Economy

Publications

Estimating Nonlinear Heterogeneous Agent Models with Neural Networks, with H. Kase and L. Melosi, Econometrica, conditionally accepted - *Link*

Learning Monetary Policy Strategies at the Effective Lower Bound with Sudden Surprises, with S. Krane and L. Melosi, Review of Economic Dynamics, 2026, Volume 62. - *Link*

Geopolitical Risk Perceptions, with Y. Bondarenko, V. Lewis and Y. Schüler, Journal of International Economics, 2024, Volume 152. - *Link*

Reversal Interest Rate and Macroprudential Policy, with M. Darracq Pariès and C. Kok, European Economic Review, 2023, Volume 159. - *Link*

Financial Crises and Shadow Banks: A Quantitative Analysis, Journal of Monetary Economics, 2023, Volume 139, p. 74-92. - *Link*

Pandemic Recessions and Contact Tracing with L. Melosi, Journal of the European Economic Association, 2023, Volume 21(6), p. 2485-2517. - *Link*

Hitting The Elusive Inflation Target, with F. Bianchi and L. Melosi, Journal of Monetary Economics, 2021, Volume 124, p. 107-122. - *Link*

Working Papers

Monetary Policy and Earnings Inequality: Inflation Dependencies, with J. Meriküll, **R&R European Economic Review** - *Link*

CBDC and Banks: Disintermediating fast and slow, with R. Bidder and T. Jackson, - *Link*

The Perils of Narrowing Fiscal Spaces, with H. Kase, L. Melosi and S. Rast - *Link*

The Macroeconomics of Stablecoins, with B. Hofmann and M. Kaldorf - *Link*

Climate Minsky Moments and Endogenous Financial Crises, with M. Kaldorf - *Link*

Geopolitical Risk in the Euro Area: Measurement and Transmission, with Y. Bondarenko, V. Lewis, N. Kang and Y. Schüler - *Link*

Coherent shocks: External identification and internal validation with an application to geopolitical risk, with Y. Schüler, S. Arndt, Y. Bondarenko, and V. Lewis - *Link*

Generative Economic Modeling, with H. Kase and F. Stohler - *Link*

Work in Progress

Nonlinear Phillips Curve and Inflation Risk, with H. Kase, L. Melosi and S. Rast

Other Publications

AI and the global economy: implications for central banks with I. Aldasoro, L. Gambacorta, E. Kharroubi; BIS Bulletin, No. 130, 28 July 2026 - *Link*

Carbon-intensive energy prices and monetary policy: lessons from a model comparison with M. Burgert, M. Darraq Pariès, L. Durand, M. Gonzalés, R. Priftis, O. Röhe, E. Silgado-Goméz, N. Stähler, and J. Varga; SUERF Policy Brief, No 1528, July 2026 - *Link*

The impact of geopolitical risk on the euro area economy: Past experience and future prospects with Y. Bondarenko, N. Kang, V. Lewis and Y. Schüler, VoxEU Article, 29 May 2026 - *Link*

Evolving approaches to monetary policy communication in the face of uncertainty: fan charts, scenarios and guidance with S. Bell, M. Chavaz, B. Hofmann, D. Rees, BIS Quarterly Review, March 2026 - *Link*

The BIS multisector model: a multi-country environment for macroeconomic analysis with M. Burgert, G. Cornelli, B. Erik, B. Mojon, D. Rees, H. Zhao, BIS Working Papers 1297, October 2025 - *Link*

A multi-sector assessment of the macroeconomic effects of tariffs with M. Burgert, B. Mojon, D. Rees, H. Zhao, BIS Quarterly Review, September 2025 - *Link*

Macroeconomic effects of carbon-intensive energy price changes: A model comparison with M. Burgert, M. Darraq Pariès, L. Durand, M. Gonzalés, R. Priftis, O. Röhe, E. Silgado-Goméz, N. Stähler, J. Varga, NGFS (Network for Greening the Financial System) Occasional Paper, September 2025 - *Link*

Geopolitical risk in the euro area with Y. Bondarenko, V. Lewis and Y. Schüler, VoxEU Article, 24 June 2025 - *Link*

Climate Minsky moments: Why they may matter less than expected with M. Kaldorf; VoxEU Article, 5 Feb 2025 - *Link*

Commodity prices and monetary policy: old and new challenges with F. Avalos, R. Banerjee, M. Burgert, B. Hofmann and C. Manea; BIS Bulletin, No. 96, 8 January 2025 - *Link*

The digital euro can strengthen financial stability, with limits with R. Bidder and T. Jackson; VoxEU Article, 3 July 2024 - *Link*

Will the digital euro strengthen financial stability? Yes, within certain limits with R. Bidder and T. Jackson; Deutsche Bundesbank Research Brief, 26 June 2024 - *Link*

Measuring geopolitical risk: Perceptions matter with Y. Bondarenko, V. Lewis and Y. Schüler; VoxEU Article, 5 June 2023 - *Link*

Estimating Growth-at-Risk: Insights from a Structural Nonlinear Model; SUERF Policy Brief, No 378, July 2022 - *Link*

The Role of Contact Tracing in the Long Pandemic War with L. Melosi; SUERF Policy Brief, No 242, Dec. 2021 - *Link*

The reversal interest rate: A new motive for countercyclical macroprudential policy with M. Darracq Pariès and C. Kok; VoxEU Article, 2 May 2021 - *Link*

In Support of Monetary Policy: Using the Countercyclical Capital Buffer to Avoid a Reversal Interest Rate with M. Darracq Pariès and C. Kok; SUERF Policy Brief, No 37, Nov. 2020 - *Link*

Enhancing macroprudential space when interest rates are "low for long" with M. Darracq Pariès and C. Kok; ECB Macroprudential Bulletin, Issue 11, 2020 - *Link*

A macroprudential perspective on replenishing capital buffers with K. Budnik, M. Darracq Pariès, C. Kok, J. Lang, M. Lo Duca, E. Rancoita, C. Rodriguez d'Acri and E. Ryan; ECB Financial Stability Review, Vol. 2, 2020 - *Link*

Teaching

04/2026	Deep Learning, neural networks, and generative modeling in macro (mini-course), European Commission, Instructor
04/2025 - 08/2025	Monetary Economics and Numerical Methods (Graduate), University of Erlangen-Nürnberg, Instructor
04/2023 - 08/2023	Monetary Economics and Numerical Methods (Graduate), University of Erlangen-Nürnberg, Instructor
09/2019 - 09/2019	Macro-Prudential Policy: A Quantitative Approach (Graduate), Florence School of Banking and Finance, TA for Enrique Mendoza
11/2017 - 01/2018	Macroeconomics I (Graduate), EUI, TA for Axelle Ferrière
10/2011 - 03/2014	Statistics (Undergraduate) , University of Erlangen-Nürnberg, TA

Scholarships

2021	PhD Grant, European University Institute
2016 - 2021	PhD Scholarship, German Academic Exchange Service (DAAD)
2019	U.S. Department Visiting Grant, European University Institute

Referee and Other Services

Referee: Review of Economic Studies, American Economic Journal: Macroeconomics, Journal of Monetary Economics, Journal of International Economics, European Economic Review, International Economic Review, Journal of Money, Credit and Banking, Journal of Economic Dynamics and Control, Oxford Bulletin of Economics and Statistics, Journal of International Money and Finance, Macroeconomic Dynamics, B.E. Journal of Macroeconomics, Economic Modelling

Scientific committee member: 4th Joint ESCB RCCC & NGFS ENR Workshop (2025), NGFS 4th Expert Network Research Workshop (2025), Deutsche Bundesbank's annual research conference (2024), ESCB Research Cluster 2 Annual Workshop (2021 & 2022)

Presentations (incl. scheduled)

2026	European Commission, Oesterreichische Nationalbank, Swiss National Bank, CEBRA 2026 Annual Meeting, 2026 IJCB Conference, TU Wien, CEPR RPN "Fintech and Digital Currencies" Annual Meeting 2026, 3rd CEPR-SAFE Frankfurt Hub Conference of RPNs
2025	Banque de France AI Methods Conference 2025, Inflation: Drivers and Dynamics 2025 Conference by ECB and Cleveland Fed, CEBRA 2025 Annual Meeting, ECB Conference on Financial Stability and Macroprudential Policy 2025, 3nd annual Non-Linearities in Macro Workshop by Bank of England, Queen Mary University of London & Lancaster University, Central Bank of Morocco, 2nd QRFE Workshop on Quantitative Finance, BSP-BIS Workshop on Macroeconomic Models, Joint Research Centre (JRC) Workshop on Energy Transition and Economic Decarbonisation, 4th Joint ESCB RCCC & NGFS ENR Workshop, ECB Working Group on Econometric Modelling, European Central Bank DG-Economics, Bank for International Settlements, University of St. Gallen

2024	University of Copenhagen, University of Zurich, Danmarks Nationalbank, ECB Banking Supervision Research Conference, KIT Quant Macro Day, European Commission Annual Research Conference (ARC) 2024, SNB Annual Research Conference, 5th Mannheim Workshop in Quantitative Macroeconomics, Bank for International Settlements, Eurosystem workshop: Digital euro & its holding limit, NGFS Research Webinar, ESCB Research Cluster on Monetary Economics, Czech National Bank Workshop on Financial Stability and Macroprudential Policy, Green Finance Research Advances 2024, Bank for International Settlements, Bank of Estonia
2023	NBER Summer Institute, University of Bonn, Bank of Finland, Bank of Lithuania, Bank of Estonia, IWH Halle, IAAE 2023, 26th Theories and Methods in Macro (T2M), EEA-ESEM 2023
2022	Banque de France, Schumpeter-BSE-Seminar (Berlin), ECB Working Group on Econometric Modelling - "Macro-at-Risk" Group, Ghent University Workshop on Empirical Macroeconomics, CEBRA 2022 Annual Meeting, 4th EUI Alumni Conference in Economics, Conference on advanced analytics: new methods and applications for macroeconomic policy (jointly organized by ECB, BoE and DAFM at King's College London)
2021	8th Conference on New Developments in Business Cycle Analysis by Danmarks Nationalbank, Bundesbank, Norges Bank, ESCB Research Cluster 1 and 2 (discussant), SNB Research Conference 2021, IAAE 2021 Annual Conference, SEC 27th International Conference, QCGBF Annual Conference 2021, Federal Reserve Bank of Chicago, Boston College, Tartu University, Bank of Canada, University of Konstanz, Bank of England, Deutsche Bundesbank, University of Tübingen, Banque de France, KU Leuven, Federal Reserve Board
2020	Norges Bank, European Central Bank DG-Research, De Nederlandsche Bank 23rd Annual Research Conference, CEPR and Bank of Finland Joint Conference on Monetary Policy Tools, VfS Annual Conference 2020, 28th Annual Symposium of the Society for Nonlinear Dynamics and Econometrics, 4rd Annual Workshop of ESCB Research Cluster 3 (discussant), European Central Bank DG-MF, NBER SI 2020 Monetary Economics (co-author presented), Danmarks Nationalbank
2019	Northwestern University, Bank of Estonia Christmas Seminar
2018	Deutsche Bundesbank, Bank of Estonia Christmas Seminar